

Hellenic Complex Systems Laboratory

Exact Confidence Intervals for a Single Proportion

Technical Report XVI

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WOLFRAM Demonstrations Project

Exact Confidence Intervals for a Single Proportion

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Search Terms: proportion, confidence interval, exact method, F distribution, inference

Short Description of the Demonstration

This Demonstration provides point estimates and exact confidence intervals for single population proportions defined by the presence or absence of a specified condition or trait. It also plots the corresponding lower and upper confidence limits as functions of the selected confidence level.

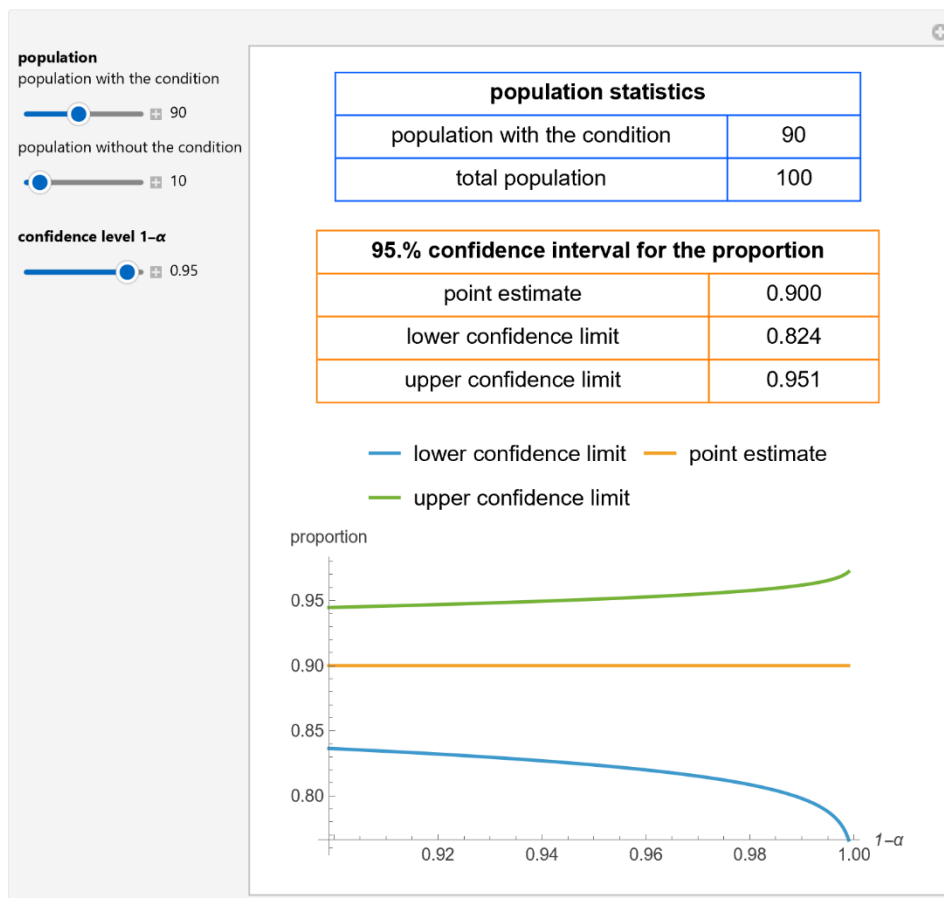


Figure 1: Population statistics, point estimate, and 95% confidence interval for a single proportion of a population obeying a condition, as well as plots of the confidence limits versus confidence level, with the settings shown at the left.

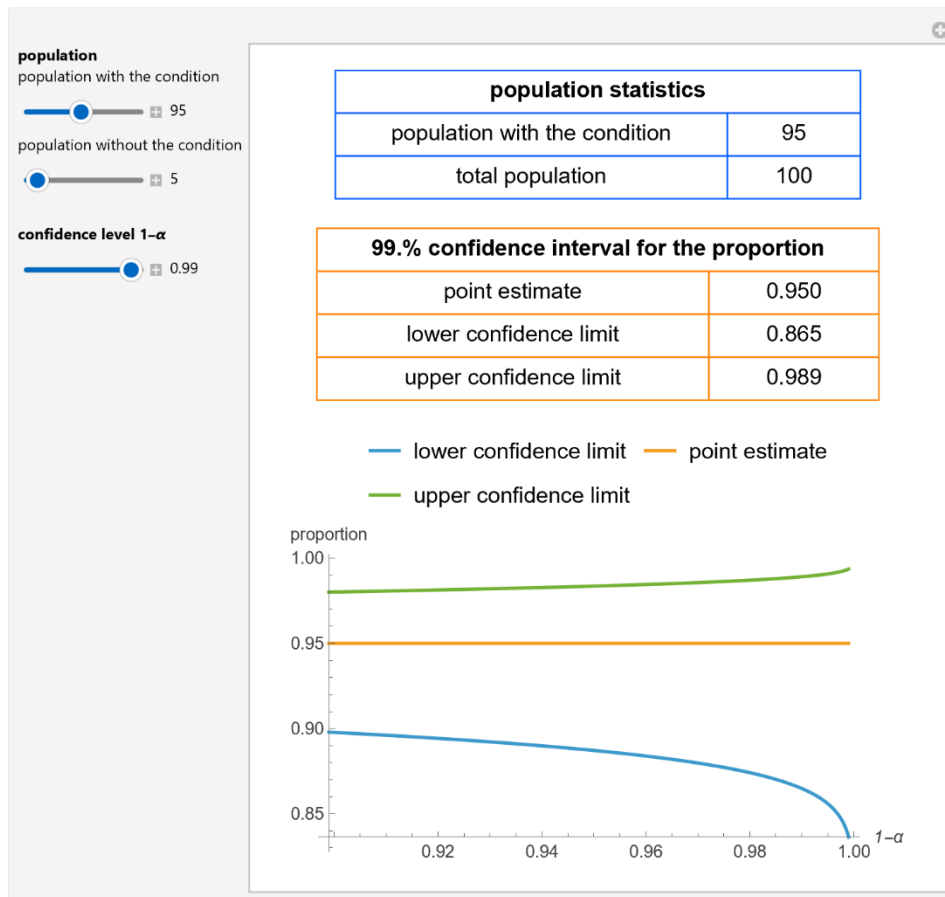


Figure 2: Population statistics, point estimate, and 99% confidence interval for a single proportion of a population obeying a condition, as well as plots of the confidence limits versus confidence level, with the settings shown at the left.

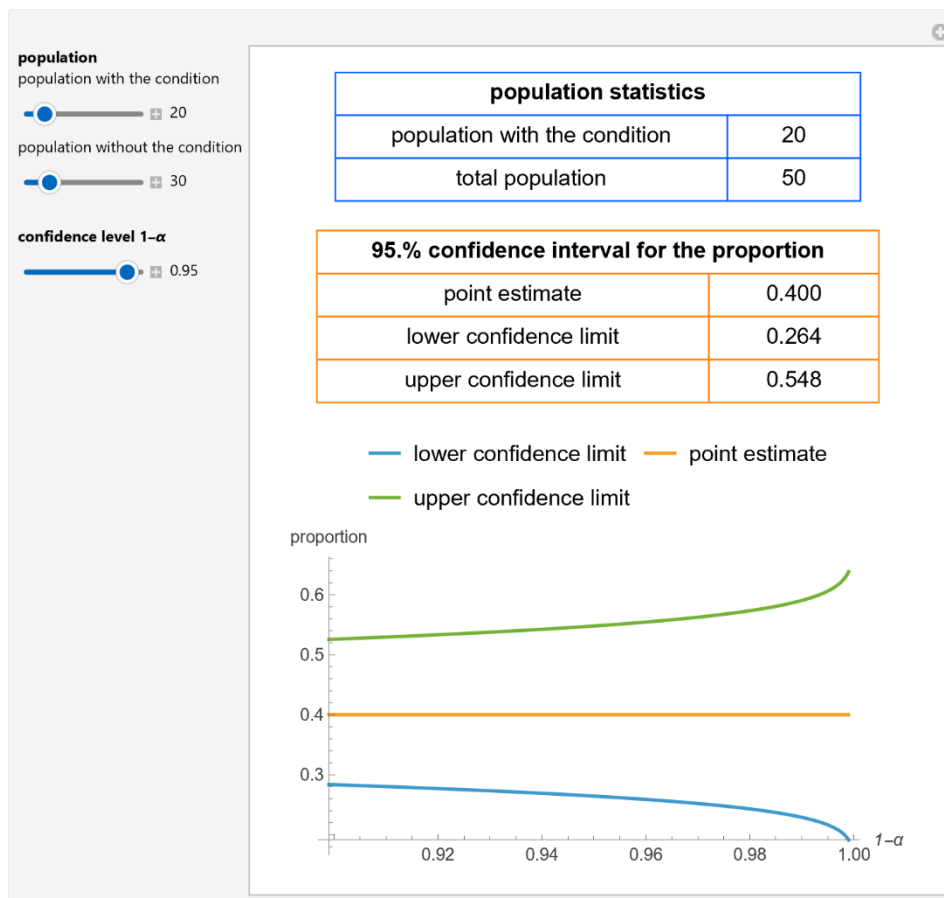


Figure 3: Population statistics, point estimate, and 95% confidence interval for a single proportion of a population obeying a condition, as well as plots of the confidence limits versus confidence level, with the settings shown at the left.

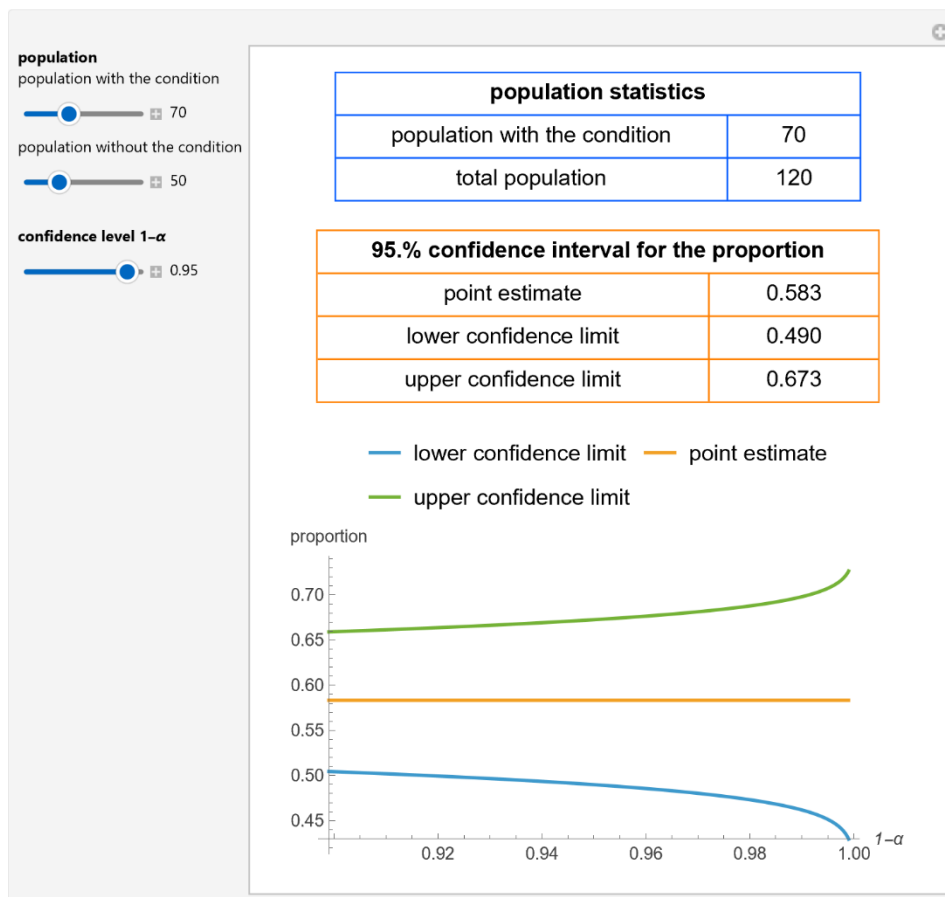


Figure 4: Population statistics, point estimate, and 99% confidence interval for a single proportion of a population obeying a condition, as well as plots of the confidence limits versus confidence level, with the settings shown at the left.

Details

Confidence intervals for individual proportions at a confidence level $1-\alpha$ are calculated using the exact method based on the F-distribution [1].

Reference

[1] J. L. Fleiss, B. Levin and M. C. Paik. Statistical Methods for Rates and Proportions, 3rd ed., Hoboken, NJ: J. Wiley, 2003.

Source

Programming language: Wolfram Language

Availability: The updated source notebook is available at:

<https://www.hcsl.com/Tools/Demonstrations/ExactConfidenceIntervalsForASingleProportion.nb>

Software Requirements

Operating systems: Microsoft Windows, Linux, Apple macOS and iOS

Other software requirements: Wolfram Player®, freely available at: <https://www.wolfram.com/player/> or Wolfram Mathematica®.

System Requirements

Processor: x86-64 compatible CPU.

System memory (RAM): 4 GB+ recommended.

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